

LD11 Variable and Method Flow

Detailed data, variable, model, metric, and search-method flow.

Field	Value
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Topic	Learning Day 11 - Hyperparameter Tuning
Document type	Individual PDF documentation
Date	2026-06-14

Submission note

This is an individual PDF document. The documentation package intentionally does not include a combined/merged PDF.

1. Why Variable Flow Matters

A notebook can run successfully and still be hard to understand if variable names and method flow are unclear. This document explains the movement from raw dataset objects to train/test splits, pipelines, search objects, predictions, metrics, and final comparison tables.

2. Dataset and Split Flow

Step	Variables	Meaning
Load classification data	classification_data, X_class, y_class	Breast Cancer Wisconsin features and target.
Load regression data	regression_data, X_reg, y_reg	Diabetes features and continuous target.
Classification split	X_class_train, X_class_test, y_class_train, y_class_test	Final train/test split for classification evaluation.
HPO validation split	X_hpo_train, X_hpo_val, y_hpo_train, y_hpo_val	Inner validation split for custom Hyperopt/Optuna style searches.
Regression split	X_reg_train, X_reg_test, y_reg_train, y_reg_test	Train/test split for Ridge Regression evaluation.

3. Model and Search Object Flow

Topic	Pipeline / search object	Output variable	Final use
KNN	Pipeline(StandardScaler, KNeighborsClassifier) + GridSearchCV	knn_search, knn_results	Classical distance-based tuning result.
Logistic Regression	Pipeline(StandardScaler, LogisticRegression) + GridSearchCV	logistic_search, logistic_results	Best explainable classification result.
Ridge	Pipeline(StandardScaler, Ridge) + GridSearchCV	ridge_search, ridge_results	Regression tuning result.
CASH	Loop over candidate model pipelines	cash_results, best_cash_model	Algorithm family selection.
Hyperopt	Objective function + search space	hyperopt_results	Advanced RF tuning result.
Optuna	Study + objective function	optuna_results	Advanced RF tuning result.
Successive Halving	HalvingRandomSearchCV or fallback	halving_results	Multi-fidelity result.
Final comparison	pd.concat of result tables	final_classification_comparison	Executive result table and chart.

4. Metric Flow

Classification models produce class probabilities and class predictions. Probabilities are used for ROC-AUC because ROC-AUC measures ranking quality across thresholds. Class predictions at the default threshold are used for accuracy and F1-score. Regression predictions are compared against continuous test targets using RMSE, MAE, and R2.

Flow takeaway

Each section follows the same logic: introduce topic, build model/search, run tuning, evaluate on test data, write output-based conclusion.